



EQUIPMENT REPLACEMENT ESTIMATE

RTU-1 / PKG-1 · PACKAGED GAS/ELECTRIC · ARPEC TRAINING FACILITY

CURIOUS MECHANICAL
SERVICES

29 SEPTEMBER 2026

This estimate covers replacing the packaged gas/electric unit designated RTU-1 in the field record and PKG-1 in the inspection report, and rebuilding the base it stands on. It is built from the bottom up, one line at a time. Where a line cannot be quoted yet it is marked as an allowance and its basis is printed, so you can see which numbers are firm and which will move. Nothing here is a range.

1 THE UNIT BEING REPLACED

NAMEPLATE, DECODED AGAINST LENNOX LITERATURE

CODE	POSITION	WHAT IT MEANS
L	Product line	L Series
G	Type	Packaged gas heat with electric cooling
A	Major design sequence	1st generation
048	Nominal capacity	4 tons. Rated net 48,000 Btu/h in the 2001 service literature; 49,000 Btu/h in the Jul 2008 bulletin, after Lennox withdrew its sub-13-SEER models
H	Cooling efficiency	High
2	Refrigerant	R-22 — not a count of compressor stages
B	Blower	Belt drive — not the gas heat code
H	Heating	High gas heat, 2 stage. 92,000 / 125,000 Btu/h input
3	Minor design sequence	3rd revision
Y	Power	208/230 V, 3 phase, 60 Hz

ONE COMPRESSOR, ONE CIRCUIT, SINGLE-STAGE COOLING

The service manual for this family states it directly: **"LGA/LCA units utilize one compressor"**, and **"all LGA/LCA 3 through 6 ton high efficiency units use one scroll compressor"**. The main control module A55 controls one compressor, one gas valve, one outdoor fan and one blower. The published data agrees throughout: one compressor RLA row, one factory charge of 9 lb 8 oz, one TXV, one drain, one crankcase heater, one contactor. The heating is two-stage; the cooling is not.

Lennox Service Literature Corp. 9822-L12 rev 09/2001, pp. 1, 18, 19. Compressor make and model are not derivable from literature and must be read off the compressor nameplate in the field.

2 WHAT WE MEASURED, AGAINST WHAT LENNOX PUBLISHES

FIELD INSPECTION SEPTEMBER 2026

READING	MEASURED	LENNOX TARGET	DEVIATION	REFERENCE
Suction pressure	66.9 psig	78.5 psig at 90 °F entering air, tolerance ±5	11.6 psig below target, 6.6 below the bottom of the band	Table 14, LGA/LCA048H
Subcooling	4.7 °F	10 °F ±1	5.3 °F below target, 4.3 below the band	Table 20, LGA/LCA048H
Superheat	39.2 °F	no published target	Reported as measured. Lennox charges this unit by subcooling and the pressure tables; the word superheat does not appear in the manual, and the metering device is a balanced-port TXV	—

The unit is low on charge. The suction table is indexed to outdoor coil entering air temperature and runs from 76 psig at 65 °F to 82 psig at 115 °F, so the comparison has to be made at the ambient on the day rather than

against a single figure. Alongside the charge, the inspection recorded rust in the gas heating section, a dirty condenser coil, air filters completely blocked, rust on the cabinet, base and stand, water ponding under the unit, and a condensate trap fitted backwards, which we corrected on site.

3 WHY WE RECOMMEND REPLACING IT RATHER THAN REPAIRING IT

- **Built October 2003.** Twenty-two years and eleven months old at the date of this estimate.
- **R-22.** No longer produced. Every charge now comes from a reclaim-only market, which is a supply and price-volatility exposure on any large loss of charge, not a shock figure per pound — we are not going to quote you one, because the numbers in circulation for R-22 disagree with each other by a factor of ten.
- **The corrosion is not in one place.** The heating section, the cabinet, the base and the stand are all rusting, and the water that is doing it is still standing under the unit. Replacing the compressor would warrant one component inside that.
- **Nothing in the cabinet carries warranty.** The original Lennox terms — five years on the compressor, ten on the aluminized heat exchanger, one on everything else — ran out between 2004 and 2013.
- **The heat exchanger has not been cleared.** Rust in the gas section means a carbon monoxide test, and that test is the one finding on this unit that is a safety question rather than an economic one. It stays open until a reading closes it.

ON SERVICE LIFE, SAID ACCURATELY

The figure usually quoted — fifteen years for a rooftop unit — traces to Akalin (1978), and ASHRAE itself footnotes that entry: the data **“may be outdated and not statistically relevant... use these data with caution”**. The newer study in the same table, Abramson et al. (2005), returns no figure for rooftop units at all for want of sample. So we are not resting this recommendation on a statistic. We are resting it on what we measured and photographed on your equipment, which is in the report, and age is context for that rather than the argument itself.

4 WHAT IS *NOT* A REASON, AND WE WILL NOT CLAIM IT

You may be told a new unit pays for itself in energy. On this machine, at this building, it does not, and the arithmetic is short enough to check.

At 1,500 equivalent full-load cooling hours and 11.37 cents per kWh, the **entire** annual cooling electricity of the existing unit is about **\$682**. That is the ceiling, and it needs no assumption at all about the new unit: a replacement that used *zero* electricity could not repay this estimate inside sixty years. Option 1 saves on the order of **\$180 a year**. The payback is longer than the life of the machine.

A claim of thirty to forty-five per cent off the electric bill would require the whole facility's annual bill to be about **\$700**. It is not. What a new unit does buy is reliability, a heat exchanger that has been tested, a refrigerant you can still buy, warranty, and humidity control the old machine can no longer hold. We would rather sell you those than a payback we cannot defend.

5 THE TWO OPTIONS

4 TON, 208/230 V 3 PHASE, GAS/ELECTRIC

	OPTION 1 — HIGH EFFICIENCY, TWO-STAGE	OPTION 2 — HIGHEST EFFICIENCY BUILT AT 4 TONS
Unit	Lennox LGT048H5E — Enlight	Lennox LGM048U5E — Model L
Rated efficiency	17.3 SEER2	19.9 SEER2 / 13.2 EER2
Compressor	Two-stage scroll	Variable-capacity inverter scroll
Refrigerant	R-454B	R-454B
Operating weight	629 lb	636 lb
Equipment price	\$7,400 allowance	\$9,500 allowance
Published data	Lennox Engineering Handbook 211109, Aug 2026	Lennox Engineering Handbook 211134, Jul 2025
Existing unit, for comparison	Rated 13.25 SEER / 11.30 EER in the 2008 bulletin, or 12.0 SEER / 10.4 EER in the 2001 literature that matches an October 2003 build. 720 lb net, 738 lb as configured. Both replacements are lighter than what is on the stand today.	

Neither option is the cheapest thing that would pass. The federal floor for this class is 13.4 SEER2 and we are not quoting it. Both options are R-454B, which is what this class is manufactured with now: the 700-GWP limit on new manufacture took effect 1 January 2025, and on a self-contained outdoor packaged unit the A2L leak detection is factory-fitted and already in the equipment price, so it adds no field scope.

BEFORE YOU BUY THE INVERTER

Option 2's variable-capacity compressor earns its money by running slowly for long periods, which is where the part-load dehumidification and the turndown come from. That needs a control system and a load that can ask for it. If this unit serves a single training bay on a simple two-stage thermostat, a good part of what you pay for is never exercised. The step from Option 1 to Option 2 is worth roughly **\$60 a year** in electricity at this building's run hours, so it should be bought for headroom, humidity control and turndown — or not bought. We will tell you which after we see how the space is controlled.

6 BOTH OPTIONS, PRICED LINE BY LINE

THE SAME SCOPE, TWO UNITS

ITEM	OPTION 1 17.3 SEER2	OPTION 2 19.9 SEER2
EQUIPMENT AND DELIVERY		
Packaged unit, 4 ton nominal, 208/230V-3ph-60Hz, R-454B, gas/electric	\$7,400	\$9,500
Inbound freight and liftgate	\$0	\$0
Commercial programmable thermostat, 2 heat / 2 cool, and wiring	\$310	\$310
REMOVAL, REFRIGERANT RECOVERY AND DISPOSAL		
R-22 recovery to 0 in Hg, cylinder handling, freight to certified reclaimer	\$150	\$150
Disposal: C and D tipping for the demolished stand and old duct, scrap-yard trip	\$225	\$225
RIGGING AND SETTING		
Rough-terrain forklift, one day including round-trip delivery	\$525	\$525
DUCT, CONDENSATE AND GAS INTERFACE		
Duct adapter plenum and two transitions, flex connectors, insulation, weather jacket, re-support ALLOWANCE	\$2,400	\$2,400
Condensate: piped route to an approved receptor, trap correctly oriented and primed	\$350	\$350

ITEM	OPTION 1 17.3 SEER2	OPTION 2 19.9 SEER2
Gas: riser and drop replaced in kind, sediment trap, appliance shutoff, union, leak test	\$500	\$500
ELECTRICAL		
NEMA 4X disconnect, whip, fuses to bring the OCPD to at or below the new nameplate MOCP, corroded conduit and fittings	\$625	\$625
NEC 210.63 service receptacle within 25 ft, same level as the equipment	\$450	\$450
FIELD LABOUR		
Field labor, fully burdened, three mobilizations	\$5,597	\$5,597
COMMISSIONING AND DOCUMENTATION		
Instrumentation amortization and the written commissioning report	\$120	\$120
PERMITS AND FEES		
Miami-Dade mechanical permit, exact change out, all-in	\$319	\$319
Miami-Dade gas permit, separate application, all-in	\$234	\$234
Miami-Dade electrical permit, all-in	\$234	\$234
Notice of Commencement, recording and certified copy	\$8	\$8
FLORIDA SALES AND USE TAX		
Florida sales and use tax	\$853	\$979
Direct cost	\$20,299	\$22,525
Overhead and profit	\$6,559	\$6,979
Disclosed contingency, 10 %	\$2,686	\$2,950
SECTION A — CHANGE-OUT	\$29,544	\$32,454
STAND, PAD AND DRAINAGE		
Shop-fabricated hot-dip galvanized equipment stand, delivered and erected ALLOWANCE	\$2,700	\$2,700
PE-sealed wind load and anchorage design, ASCE 7 / FBC 1609, HVHZ ALLOWANCE	\$1,600	\$1,600
Concrete pad, formed and rebarred, 4000 psi, approx 48 sq ft, incl. sub mobilization ALLOWANCE	\$2,600	\$2,600
Subgrade overexcavation, geotextile and compacted stone base ALLOWANCE	\$1,300	\$1,300
Grade correction and surface drainage so water no longer ponds under the unit ALLOWANCE	\$850	\$850
Building permit for the replacement equipment stand at grade ALLOWANCE	\$352	\$352
Direct cost	\$9,402	\$9,402
Handling and contingency	\$1,974	\$1,974
SECTION B — THE BASE	\$11,376	\$11,376
TOTAL	\$40,920	\$43,831

Option 1 totals \$40,920, or \$10,230 per ton. Option 2 totals \$43,831, or \$10,958 per ton — the whole difference between them is the unit itself, since the scope is identical. Labour is 96 man-hours over three mobilisations, billed at about \$96 an hour, which is a deliberate below-market position: a Miami-Dade public university awarded \$155.00 an hour for plant service in a comparable procurement.

7 HOW THE TWO UNITS ARE PRICED, SINCE LENNOX DOES NOT PUBLISH IT

THREE ROUTES, AND THE ARITHMETIC

Lennox publishes no commercial pricing. Every price on LennoxPros sits behind a dealer sign-in and neither model is returned by the public catalogue search — we checked again on 30 September 2026. Carrier renders price only

after login, and the upper-tier Trane models are carried by no public-price retailer we could reach. At 4 tons on three-phase power, no packaged gas/electric unit above 16.4 SEER2 has a purchase price obtainable outside a distributor account.

So we priced them the way an estimator prices anything without a catalogue: from equipment we **can** price, by three independent routes, and the arithmetic is printed below so you can check it. Both figures are declared allowances and both are trued up against the distributor quote before anything is ordered.

OPTION 1 — 17.3 SEER2, TWO-STAGE

ROUTE	BUILT FROM	RESULT
1 — premium brand	Trane YSK048A3 at \$6,909 , a verified R-454B 4-ton three-phase unit, plus \$705 of two-stage content (the measured Daikin single-to-two-stage delta), plus 1.2 SEER2 points at \$261 a point to reach 17.3	\$7,927
2 — value brand	ACIQ R-454B code-minimum at \$6,050 , the same \$705 of two-stage content, plus \$313 of efficiency	\$7,068
3 — architecture match	Daikin DHG0483DM at \$5,537 — a verified 16.4 SEER2 <i>two-stage</i> 4-ton three-phase unit, so it matches the architecture and not only the size — times 1.20 for R-454B, plus 0.9 SEER2 points	\$6,879
Weighted 40 % / 35 % / 25 %, favouring the architecture match, then set at the round figure for a channel with no discount route		\$7,400

OPTION 2 — 19.9 SEER2, VARIABLE-CAPACITY INVERTER

The inverter step is added in **dollars, not as a percentage**. Across 3, 4 and 5 tons the measured step is almost flat — \$1,598, \$1,779, \$1,600 — which is the signature of fixed content, a drive board and an inverter compressor, rather than of a ratio that scales with price.

MEASUREMENT	BUILT FROM	ADDER
In class, packaged	A packaged R-454B pair, corrected so the two-stage step already bought in Option 1 is not charged twice	\$1,100
Same tonnage, split	A 4-ton two-stage to variable-speed pair, priced at one vendor on one day	\$1,904
Midpoint, uplifted about 14 % for the commercial channel, plus \$350 for the three-phase drive and for reaching the top of the class		\$2,100
Option 1 at \$7,400, plus the adder		\$9,500

WHICH WAY THESE COULD MOVE

Not one way. Upward: the commercial channel has no discount route, and the cost of efficiency climbs faster near the top of a range than in the middle. Downward: two of the three routes borrow a step measured on residential equipment, a 4-ton unit sits at the bottom of a 3-to-25-ton family where prices compress, and the gas heat input is not yet fixed — one heat-size step is worth about \$220 either way. We have not leaned the number in either direction.

A sanity check that costs nothing: Option 2 works out at about \$2,375 a ton. A verified premium-brand code-minimum unit is \$1,727 a ton, and semi-custom equipment runs well past \$3,000. It sits where it should.

WHAT CLOSES BOTH NUMBERS

One written quote from a Lennox commercial distributor serving Miami-Dade, listing **LGT048H5E** and **LGM048U5E** at 208/230 V three phase side by side, with the gas heat input we select from a Manual N load calculation rather than inherited from the existing 92/125 MBH, which is oversized for this climate.

Every dollar the quote lands away from these allowances moves the total by **\$1.39** — markup, tax and contingency included. You can do that arithmetic yourself the day the quote arrives.

8 ALLOWANCES, CONDITIONAL WORK AND WHAT WE COULD NOT PRICE

DECLARED ALLOWANCES INSIDE THE TOTAL

LINE	WHY IT IS AN ALLOWANCE AND WHAT CLOSES IT
Shop-fabricated hot-dip galvanized equipment stand, delivered and erected	BUDGET ALLOWANCE, estimator judgment, NOT a quote. Takeoff assumes about 350 lb: an 86 x 45 in perimeter frame plus two cross members in L3×3×1/4 or HSS3×3×1/4, six posts at 24 in, six 6×6×3/8 base plates. A per-pound extension is deliberately NOT used: tonnage rates presume a production run, and a one-off stand is governed by the misc-metals shop minimum job charge and the galvanizer minimum lot charge. Mill steel runs about 1.20 per lb, fabrication adds 0.20 to 0.50, galvanizing 0.40 - under 700 dollars of content, which is exactly why the minimums and not the content set the price. Priced ONCE, as a sub: no CMS erection hours are carried against it.
PE-sealed wind load and anchorage design, ASCE 7 / FBC 1609, HVHZ	BUDGET ALLOWANCE, estimator judgment, NOT a quote. No Florida firm publishes a fee for this scope. Engineering Express, the firm that sealed the MIRO stand NOA, publishes 80 to 100 dollars only for pre-engineered catalog tie-down plans that anchor a unit to an existing compliant substrate - those do NOT design a 24 in elevated stand, and its custom work routes to a private quote. The lead time sits AHEAD of the pour in the schedule, not after it.
Concrete pad, formed and rebarred, 4000 psi, approx 48 sq ft, incl. sub mobilization	BUDGET ALLOWANCE, estimator judgment, NOT a quote. At roughly 0.6 cubic yards this is far below a ready-mix load minimum, so the pour is governed by short-load and mobilization charges plus a return trip to strip forms, not by any per-square-foot rate. Required under either stand route.
Subgrade overexcavation, geotextile and compacted stone base	BUDGET ALLOWANCE, estimator judgment, NOT a quote. Water ponds under the unit today, so the subgrade is unsuitable as found and is not inside any pad rate.
Grade correction and surface drainage so water no longer ponds under the unit	BUDGET ALLOWANCE, estimator judgment, NOT a quote. THE ROOT-CAUSE LINE. Surface drainage is designed first. A dry well is NOT assumed: South Florida seasonal groundwater may defeat it and no boring has been taken at this site.
Duct adapter plenum and two transitions, flex connectors, insulation, weather jacket, re-support	ALLOWANCE. Supply and return opening sizes AND positions differ between the 2003 cabinet and every 2026 candidate, and the unit moves onto a stand of different height. Held open until the existing openings are measured and the selected unit's certified dimensional submittal is in hand.
Building permit for the replacement equipment stand at grade	NO product approval or NOA is required: F.S. 553.842(5) enumerates the categories subject to statewide product approval and mechanical equipment stands are not among them. What IS required is a building permit, taken as IO 4-63 sec. I.B.5 Alterations and Repairs, whose 254.40 minimum governs a 28 sq ft footprint. Sits in Section B because it is the stand, not the change-out.

SECTION B IS THE WEAKEST NUMBER ON THIS PAGE, AND WE ARE SAYING SO

Every line in the base section is estimator judgement, not a quote. Two quotes close it: a misc-metals fabricator and a concrete subcontractor, each walking the site. We deliberately did **not** price the stand by the pound. Per-pound steel rates presume a production run; a single one-off stand is governed by the fabricator's minimum job charge and the galvanizer's minimum lot charge, and pricing it by the pound is wrong in kind rather than just in degree — it is how an estimate ends up below what any subcontractor would honour.

We also checked whether a catalogue stand with a Miami-Dade product approval would avoid the engineering fee. It will not. The relevant approval covers units up to 48 in wide at 400 lb; your cabinet is 86 in at about 640 lb, and the sealed drawing states that any dimension beyond the listed values requires site-specific certification. It would not have avoided the concrete either, because it anchors into a 3,000 psi substrate. No product approval is required for the stand itself — equipment supports are not among the categories the statute lists — but a building permit is, and it is carried.

CONDITIONAL, PRICED BUT NOT IN THE TOTAL

ITEM	TRIGGER	IF IT APPLIES
licensed electrician and 3-pole breaker if the	Priced ONCE, as a conditional allowance. Base scope assumes a local fused disconnect and the mechanical crew's hours. Determined by inspection before the	\$895

ITEM	TRIGGER	IF IT APPLIES
overcurrent device is at a remote panel	order, not after.	
temporary spot cooling, one week, 5 ton spot cooler with round-trip mobilisation	Only if the pad cannot be poured ADJACENT to the existing stand. Pouring adjacent lets the unit shift sideways with no cooling outage and is the cheaper, lower-risk answer. Needs the Lennox service clearance confirmed on all four sides.	\$1,200

On the sequence: the new pad occupies the ground the old unit stands on, so if it is poured in place the building loses cooling for the cure. We intend to pour it **alongside** the existing stand and shift the unit sideways, which costs nothing extra and keeps you running. That needs the service clearance confirmed on all four sides before we commit to it; if the clearance is not there, the temporary cooling line above applies and its power supply is single phase, so it cannot reuse the existing three-phase disconnect.

TWO DATED ITEMS

The permit figures are the schedule in force to 30 September 2026. Miami-Dade Implementing Order 4-63 states its fees run through that date and rise each 1 October by the increase in the Consumer Price Index, so they are to be re-checked against the current fee sheet before the applications are filed. Every permit number here is stated as of 29 September 2026.

The overcurrent device has to come down. The existing unit is 27 A minimum circuit ampacity with 40 A maximum overcurrent protection. Every 2026 candidate is lower. Leaving a 40 A device on a unit whose nameplate permits 35 would exceed the nameplate maximum, so this is a code item and not a judgement call. The existing conductors are adequate, because the ampacity falls.

9 EXCLUDED FROM THIS ESTIMATE

- Anything found inside the duct, the gas riser or the electrical raceway once the old unit is off the stand, beyond the allowances named above.
- Asbestos, lead or any other hazardous material survey or abatement.
- Ductwork beyond the transitions and the adapter plenum: cleaning, sealing, re-insulation or re-sizing of the existing distribution.
- Any change to the served space: diffusers, grilles, zoning, or a second thermostat.
- An economizer. None is required — Miami-Dade is climate zone 1A and the unit is below the 54,000 Btu/h trigger — and none is included.
- A crane, a curb adapter, roofing, flashing or roof-warranty coordination. This unit is at grade on a steel stand, so no such work arises and none is carried.
- Structural redesign of anything other than the equipment stand.
- After-hours or weekend premium time. The estimate is straight time.
- A payment or performance bond, additional-insured endorsements beyond our standard certificate, and certified-payroll administration.
- Electrical utility work, service upgrade, or any change of the incoming supply.
- Balancing or commissioning of equipment other than the unit we install.

HOW THIS NUMBER CHECKS OUT, AND WHERE IT DOES NOT

An independent estimator reviewing this same scope separately, without seeing this build, put it at about **\$41,000**. This build totals **\$40,920** — a two per cent difference arrived at by two routes.

What we could **not** obtain: the RSMeans section for packaged rooftop units is behind a subscription, and no public bid tabulation for a four or five ton change-out surfaced. Figures for this do circulate on cost-aggregator websites. We are not quoting them, because on the same day those sites disagree with each other by more than the number is worth, and because a benchmark you cannot trace is not a check — it is decoration. This price is defended line by line or not at all.

